

The Influence of Metacognitive Difficulty Judgements on Persistence in Problem-Solving

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Background:

Metacognition involves **monitoring** one's own cognitive processes, which informs **control** decisions¹

Difficulty judgments (JODs) are prospective monitoring judgments about how hard a task will be^{2,3}

Studies suggest JODs are based on salient task features (i.e. **surface cues**) rather than **actual difficulty**³

JODs may influence **control decisions** such as persistence (time on task) or disengagement

Individual traits like **Grit** may influence how long people persist when tasks feel difficult^{4,5}

Hypotheses:

H1: JODs will be influenced by misleading surface cues like pronounceability; easy to pronounce anagrams will be rated as easier²

H2a: Actual difficulty cues (e.g. solvability and solution convergence) will influence persistence on task; participants will disengage from unsolvable and unlikely convergence items faster and more frequently

H2b: JODs will predict persistence and disengagement

H3: Individual higher in Grit will persist for longer and disengage less frequently⁵

Method:

Participants: N = 91 ($M_{\text{age}} = 32.6$; $M_{\text{female}} = 54.9\%$)

Design: 2 (Pronounceability) x 2 (Solution convergence) x 2 (Solvability) within-subjects

Pronounceability: OARLY vs. AOSLR

Solution convergence: TEPID vs. COMFY

Solvability: OARLY vs. MEDRY (unsolvable)

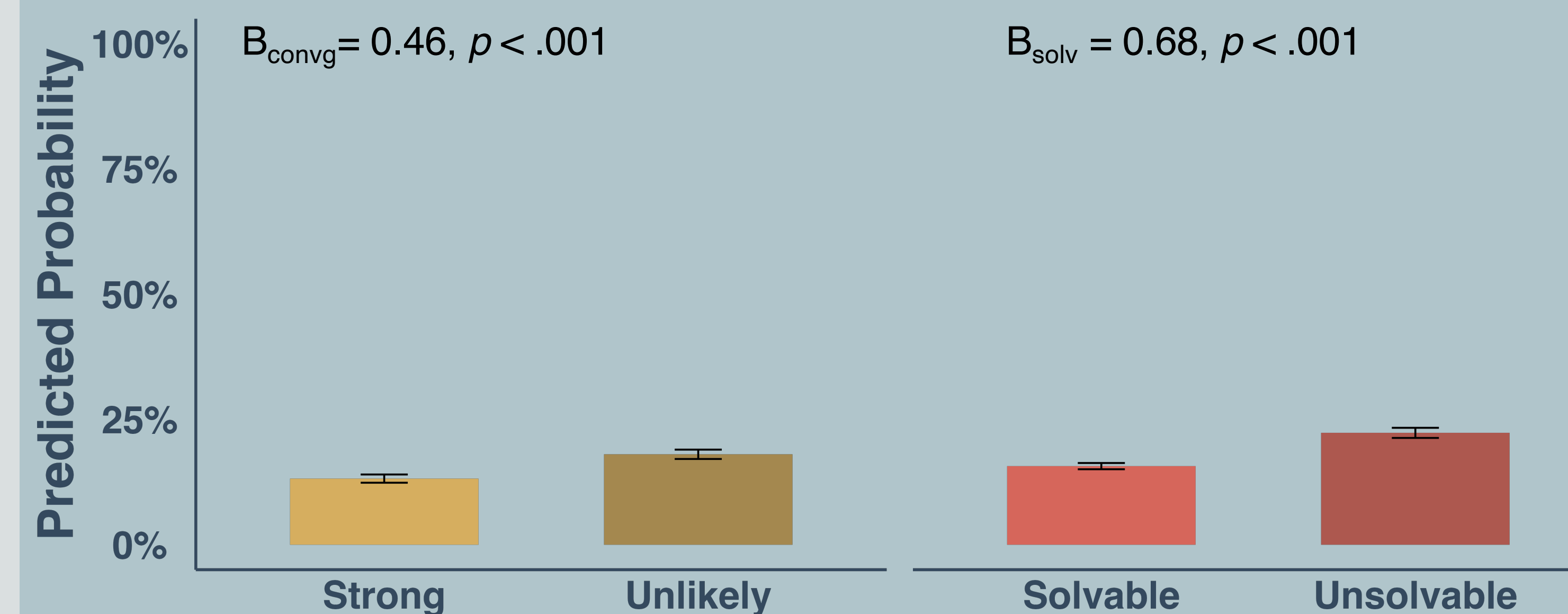
Procedure: Anagram (1s) → JOD → Anagram → Solve or Disengage (give up or classify unsolvable)

Results:

Easy to pronounce items were judged to be less difficult ($B = 2.54$, $SE = 0.99$, $p = .011$). JOD was not affected by solvability or solution convergence.

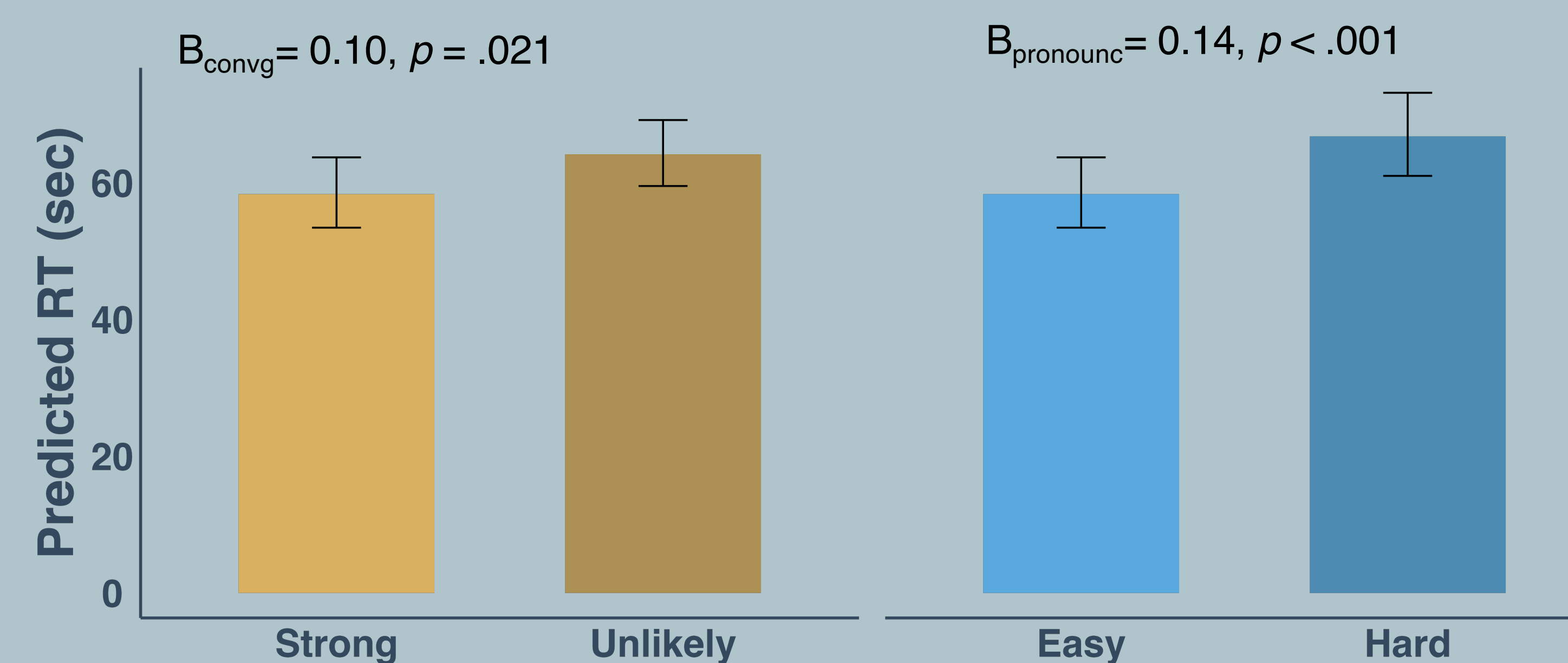
Participants gave up more frequently on unlikely convergence and unsolvable items. JOD did not predict giving up.

Give up ~ JOD + Grit+ Pronounceability + Convergence + (1|Participants) + (1|Items)
Give up ~ JOD + Grit+ Pronounceability + Solvability + (1|Participants) + (1|Items)



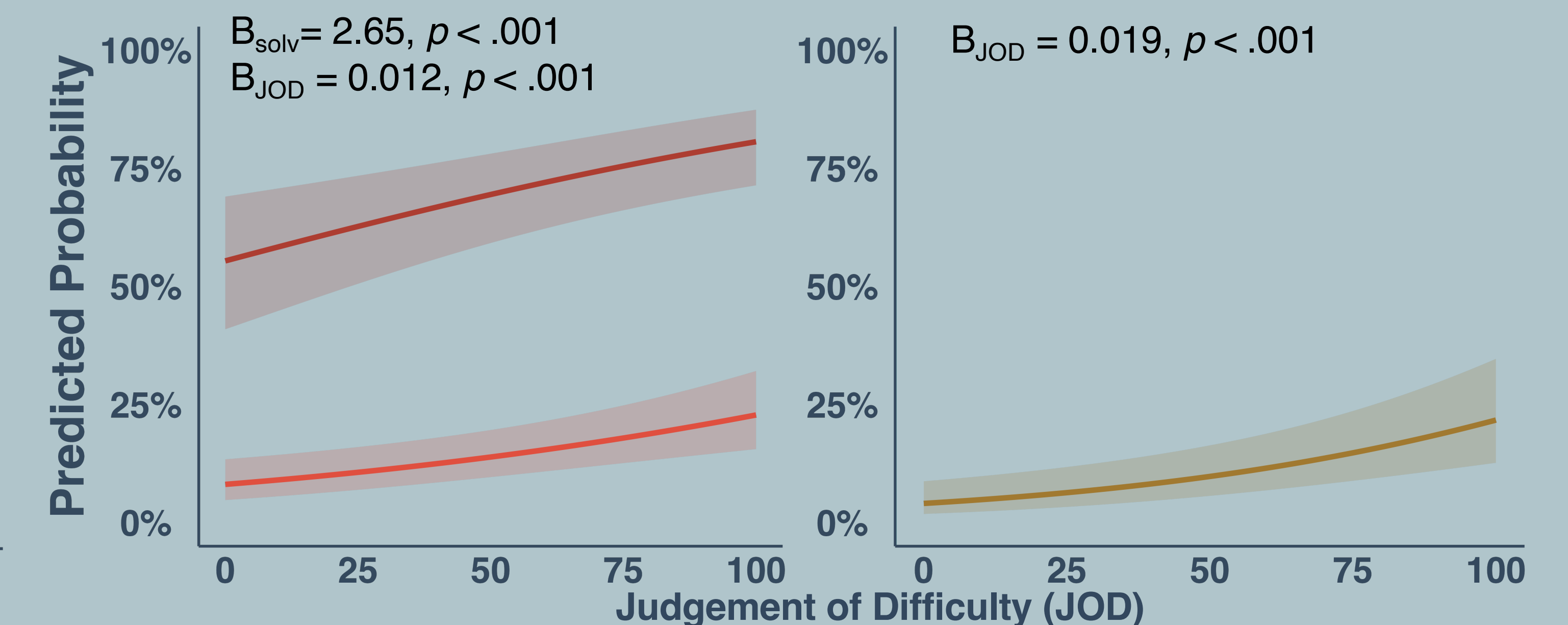
Participants took longer to give up on unlikely convergence and hard to pronounce items.

Giving up RT ~ JOD + Grit+ Pronounceability + Convergence + (1|Participants) + (1|Items)



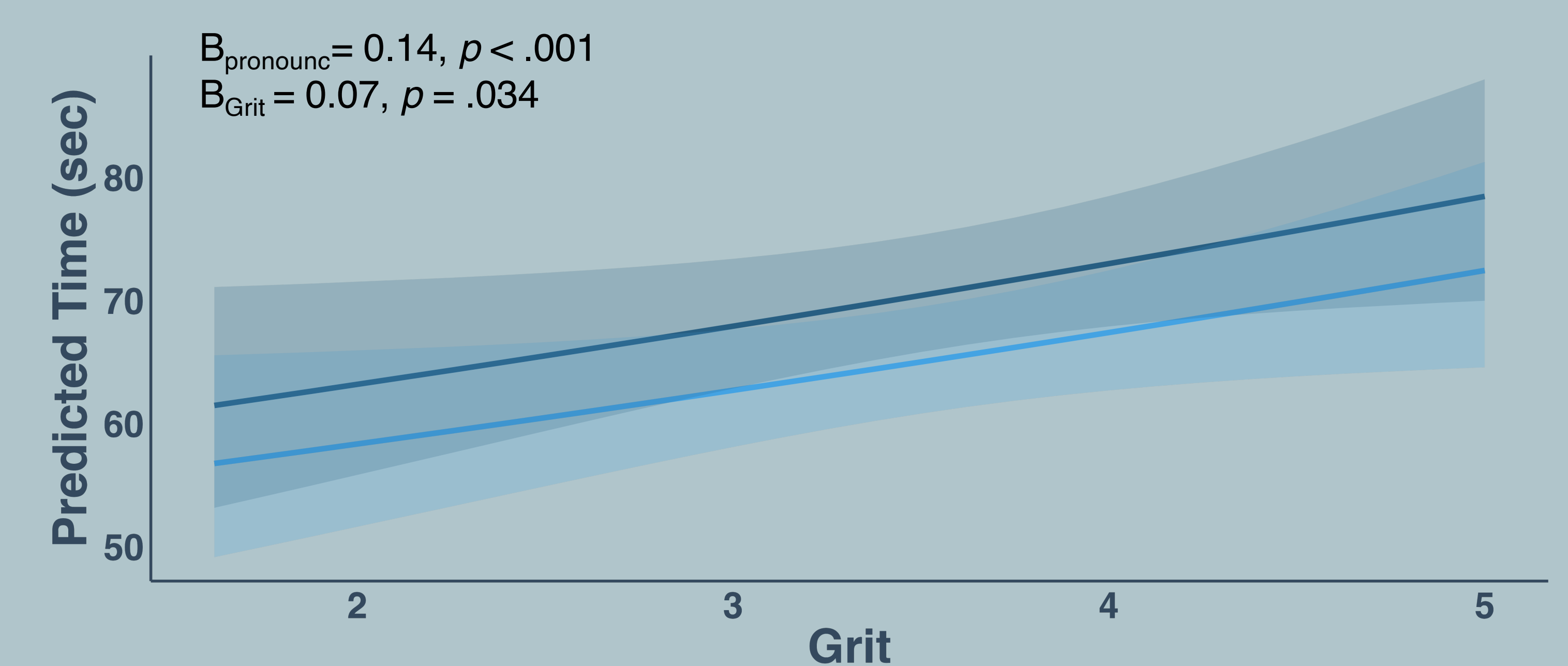
Unsolvable items are more likely to be classified as unsolvable; higher JOD predicts increased likelihood of classifying items as unsolvable.

Unsolvable ~ JOD + Grit+ Pronounceability + Solvability + (1|Participants) + (1|Items)
Unsolvable ~ JOD + Grit+ Pronounceability + Convergence + (1|Participants) + (1|Items)



Participants higher in Grit persisted longer before giving up.

Giving up RT ~ JOD + Grit+ Pronounceability + Solvability + (1|Participants) + (1|Items)



Discussion:

Surface cues influenced JOD, while cues to actual difficulty guided persistence in problem-solving:

- People disengage from harder items more frequently by either giving up or classifying items as unsolvable; but they persist longer before giving up on harder items*
- **Mixed results on the predictivity of JODs:** it predicts classifying items as unsolvable but does not predict giving up or persistence

*Note: The three mutually exclusive outcomes on each trial (solving the anagram, giving up, or classifying unsolvable) may explain counterintuitive results; easy items are solved quickly and removed from giving up analyses.

High Grit individuals persisted longer on items before giving up.

References:

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4. Duckworth, A. L., & Quinn, P. D. (2009). Development and Validation of the Short Grit Scale (Grit-S). *Journal of Personality Assessment*, 91(2), 166–174.
5. Lucas, G. M., Gratch, J., Cheng, L., & Marsella, S. (2015). When the going gets tough: Grit predicts costly perseverance. *Journal of Research in Personality*, 59, 15–22.